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MULTICENTER CLINICAL TRIALS ON A NOVEL POLYHERBAL FORMULATION IN ALLERGIC RHINITIS

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Summary: *Allergic rhinitis is the most frequently occurring immunological disorder. It affects men, women and children and represents significant cost in terms of suffering and loss of productivity. Allergy is termed as an excessive reaction to an environmental allergen. Pollen, mold, dust, mite and animal allergens that contact the nasal or eye lining cause sneezing, nasal congestion and itchy, watery, swollen, red eyes. Although a broad spectrum of therapeutic options is available, the treatment of allergic rhinitis appears to be far from satisfacto-*

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ry. A novel polyherbal formulation (PF; Aller-7/NR-A2) comprising seven medicinal herbal extracts was assessed in a multicenter clinical trial involving 545 patients (321 males and 224 females) aged 18-59 years for 12 weeks to evaluate its clinical efficacy in patients suffering from allergic rhinitis. A total of 171 patients participated in double-blind, randomized, placebo-controlled studies in three centers, while 374 patients were included in the open-label studies in 11 centers. The three major symptoms (sneezing, rhinorrhea and nasal congestion) of allergic rhinitis were significantly reduced. Significant improvement was also observed in absolute eosinophil count, mucociliary clearance time, peak expiratory flow rate and peak nasal flow rate. No serious adverse events that warranted cessation of treatment were observed. Minor adverse effects were noted in both the treatment and placebo groups. Thus, this study demonstrates that Aller-7/NR-A2 is well tolerated and efficacious in patients with allergic rhinitis.

Introduction

Allergic rhinitis, or hay fever, is the most common allergy and is manifested by itching of the nose, pharynx and larynx; sneezing; rhinorrhea; and nasal congestion. The initial stages of allergic rhinitis are seasonal, but later often become perennial. The most common allergens accounting for the majority of allergies are house dust mite, salivary proteins from cats and dogs and pollens, *etc*. The use of the term "hay fever" is often incorrect as hay is not the causative agent nor is fever a presenting symptom (1). Several potential determinants have been proposed for this phenomenon, including a lack of severe and repeated infections (2), obesity and lack of physical exercise (3), changes in dietary habits (4), increases of indoor allergen exposure (5) and possibly some other undefined factors.

Allergic rhinitis is a frequent immunological disorder with a prevalence of 1 in 10 in the USA (6). In clinical practice, allergic rhinitis constitutes 55% of all allergies in India (7), which makes it the most common chronic disease of today. The prevalence of allergic rhinitis increases until approximately the age of 20, remains at that level until about the age of 50, and shows a marked decrease above that age, although complete spontaneous remissions are uncommon (1).

The first line of defense in the treatment of allergies is avoiding exposure to allergens. Conventional pharmacotherapy includes antihistamines, mast cell stabilizers and corticosteroids. Antihistamines, such as promethazine and chlorpheniramine, have limited use in allergic rhinitis due to their sedative effect. Newer antihistamines, such as fexofenadine and cetirizine, despite claims to be free from sedation, still cause mild sedation and fatigue, while terfenadine and astemizole had to be withdrawn from the market due to serious adverse effects (8). Among the second-generation antihistamines, azelastine and levocabastine have been found to be locally effective and with minimal adverse effects on nasal congestion. Hence, these agents may be preferred, but they offer only symptomatic relief, particularly for sneezing and rhinorrhea.

A novel polyherbal formulation (PF; Aller-7/NR-A2), for which a US patent application has been examined and was allowed for issuance as a patent (9), was designed with an activity spectrum to counteract symptoms of allergic rhinitis. A wide range of studies were conducted to establish its pharmacological profile for antihistamine, antioxidant and anti-inflammatory and related activities (10-12); safety and toxicological studies (13); and clinical studies in healthy human volunteers and patients suffering from allergic rhinitis (14). At this stage, it was decided to study the effects of PF in a larger group of patients situated in different geographical parts of India.

Materials and methods

Patients. A total of 545 adult patients (321 males and 224 females) were selected from 14 trial centers, 171 patients in double-blind, randomized, placebo-controlled studies in three centers and 374 patients in the open-label studies in 11 centers. Since allergies are frequently dependent upon environmental factors, different geographical locations were chosen within India to evaluate larger representative groups with the disease, rather than just one circumscribed area of high incidence. Only subjects with established cases of allergic rhinitis of a chronic nature were included.

Table I demonstrates the demographic data as well as disease duration and seasonality. All centers agreed on a common protocol using Good Clinical Practices and adherence to the recommendations of the World Medical Association Declaration of Helsinki (15). Ethical Committee/Institutional Review Board approvals were obtained from the respective institutions prior to initiating the trials. The data were recorded in case record forms.

All patients were screened according to the inclusion and exclusion criteria listed in Table II. Only eligible patients were given an informed consent form and participation agreement, explaining the terms of the consent form and legal agreement in their native language and in the presence of a witness of the patient's choosing, and were asked if they wanted to participate in the study. If their reply was in the affirmative, they were asked to sign the consent form/legal agreement. A detailed history of the patient, covering presenting symptoms and their duration and severity; personal and family medical history; and past illnesses, treatments and drugs being taken, was then recorded. Clinical examination was conducted to evaluate their vital parameters and condition of the nasal tract and to exclude any organic cause.

Polyherbal formulation (PF). This novel PF (Aller-7/NR-A2) was obtained from InterHealth Nutraceuticals (Benicia, CA, USA/Natural Remedies Pvt. Ltd., Bangalore, India) and was prepared using a propri-

Table I *Demographic data*

	Open-label trial	Double-blind trial
Number of patients		
Male	218	103
Female	156	68
Age	31.037 ± 10.026	30.47 ± 11.064
Type of allergic rhinitis		
Seasonal	84	55
Perennial	290	116
Symptom duration		
1-2 years	(*n* = 137) 1.474 ± 0.5604	(*n* = 59) 1.478 + 0.5701
2-5 years	(*n* = 110) 3.890 ± 0.876	(*n* = 70) 3.723 ± 0.918
More than 5 years	(*n* = 127) 11.36 ± 5.498	(*n* = 42) 9.5 ± 3.416

Table II *Multicenter clinical trial inclusion and exclusion criteria*

Inclusion criteria	Exclusion criteria
18-60 years of age	Allergy to any medication
Clinically normal	Any serious systemic illness
History of allergic rhinitis for at least 1 year	Psychiatric disorders (difficulty in obtaining consent)
At least two of the three allergic rhinitis symptoms (sneezing, rhinorrhea and nasal congestion)	Regular smokers (more than 20 cigarettes daily)
Normal hematological profile	Drug dependence or chronic heavy alcohol consumption
Normal liver/renal function tests	Use of any microsomal enzyme–modifying drugs within 30 days preceding the study
Normal biochemical and microscopic examination of urine	Participation in any clinical trial within 6 weeks preceding the study
Voluntary written consent	

etary technique from the dried bark, fruits and rhizome of seven extensively used Indian medicinal plants (dried bark of *Albizia lebbeck*, dried fruits of *Terminalia chebula, T. bellerica*, *Phyllanthus emblica*, *P. nigrum* and *P. longum* and the dried rhizome of *Zingiber officinale)*. The final extract was standardized to 250 mg/g polyphenols as tannic acid, 85 mg/g chebulic acid, 50 mg/g gallic acid, 17.5 mg/g corilagin, 7.5 mg/g 1,3,6-tri-O-gallyol glucose, 4.0 mg/g ellagic acid, 1.0 mg/g piperine and a total of 0.20 mg/g of 6- and 8-gingerols (US Patent US20030194452 dated October 16, 2003).

Criteria for diagnosis. Patients with a history of allergic rhinitis for a minimum of 1 year were included in the trial. Further, at the time of inclusion, at least two out of three nasal symptoms, *i.e.*, sneezing, rhinorrhea and congestion, were required to have been present for at least 15 days prior to inclusion (6). Clinical edema of the nasal mucosa and turbinates was considered to be diagnostic (7). Skin prick test was done in a group of patients who had an association with any other allergic disorder and/or family history of allergy.

Administration of test supplement. All patients were advised to take the PF or placebo as two capsules after breakfast and two capsules after dinner. Each capsule contained 330 mg active extract, providing a daily dose 1,320 mg. Placebo capsules contained inert ingredients (lactose) and were similar in appearance to the test capsules. The patients were provided with bottles containing 60 capsules and were advised to bring the bottles back for replacement every 2 weeks, regardless of the number of remaining capsules. A capsule count of less than 15% of remaining capsules was taken as compliance with treatment.

Symptom assessment. The various symptoms assessed are described in Table III. The severity of each symptom was measured on a visual analogue scale by the patient at each visit and recorded in a case record form (7, 16). A score of 100 indicated a very severe condition, which disturbed all routine activities, and 0 indicated the absence of any symptoms whatsoever.

Biochemical parameters. Laboratory tests were carried out to exclude diseases of blood, liver and kidney following a fixed protocol (Table IV).

The physicians were given the liberty to include serum IgE measurements, X-ray of chest and paranasal sinuses, electrocardiogram, mucociliary clearance, peak nasal flow rate (PNFR) and pulmonary function tests.

Table III *Clinical symptoms assessed**

Sneezing
Nasal congestion (stuffy nose)
Rhinorrhea (runny nose)
Headache
Wheezing
Cough
Body ache
Fever
Anorexia
Other symptoms

*The duration and severity of each symptom was assessed on a visual analogue scale (0-100 mm).

Objective parameters

Absolute eosinophil count. The presence of more than 500 eosinophils/μl of blood is common in bronchial asthma, cutaneous allergic reaction, hypersensitivity states, invasive helminthic infections and allergic reactions to drugs such as iodides, aspirin and sulfonamides *etc*. (1). Allergies such as hay fever, asthma, eczema, serum sickness, allergic vasculitis and pemphigus are commonly associated with eosinophilia.

Table IV *Biochemical evaluations*

1.	Blood:
	Hemoglobin
	Differential count
	Random blood sugar
2.	Liver function:
	Serum bilirubin
	Serum glutamate oxaloacetate transaminase
	Serum glutamate pyruvate transaminase
	Serum alkaline phosphatase
	Serum protein
3.	Renal function:
	Albumin/globulin ratio
	Blood urea
	Serum creatinine
4.	Urine:
	Sugar
	Protein
	Microscope

Total serum IgE levels. Immunoglobulins (Ig) are products of differentiated B-cells and mediate the humoral arm of immunity by binding to antigen and inactivating the offending foreign substance from the body. IgE is the major type involved in allergy by arming mast cells and basophils. Total serum IgE levels were measured in four centers, with 174 patients. The measurements were done at two time points, week 0 and week 12, using a chemiluminescence method (17).

Mucociliary clearance time (MCT) (saccharin transport time [STT]). STT was used as an indicator of inflammatory changes in the nasal mucosa (18). Saccharin particles are transported along the mucosa due to ciliary action. A 1-mm diameter of saccharin particle was placed approximately 1 cm behind the anterior end of the inferior turbinate in the floor of the nasal cavity and the time required to perceive a sweet taste was noted. This was done on each side of the nasal cavity. All patients were instructed not to inhale or exhale forcefully and to refrain from any nasal manipulation and sniffing. They were instructed to swallow every 30 sec and report the first change in taste sensation immediately. To ensure the reliability of the procedure, the nature of the particle was not disclosed to the subjects.

Peak expiratory flow rate (PEFR). Allergic rhinitis is often coexistent with bronchial asthma. In some patients, symptoms of asthma may not be manifest, but lung function may be altered (19). As a part of this screening, PEFR was carried out by asking patients to inhale to their maximum capacity and then to completely exhale into a flow meter. The reading was recorded and repeated three times. The greatest of the three recordings were recorded as the PEFR.

Peak expiratory nasal flow rate (PNFR). To objectively assess the effect of PF on nasal obstruction, PNFR was measured in patients prior to and after treatment. The nasal flow rate was measured using a Wright's peak flow meter with modifications (20). The mouthpiece was connected to a mask, which fits snugly around the nose and mouth, and patients were asked to blow their noses into the mask after maximum inhalation. The greatest of three readings was recorded for evaluation. PNFR evaluations were repeated at each visit (20).

Skin prick test. Atopic status was measured by skin prick test using 22 common allergens applied to the forearm with a lancet to prick the skin (21). Histamine and glycerinated saline were used as positive and negative controls. Orthogonal wheal diameters were obtained and recorded from the positive control and any allergen responses were registered. The histamine rate was calculated as the ratio of allergen wheal diameter compared with the histamine wheal diameter. Results were considered positive when the histamine rate was ≥ 0.5.

Adverse events. Adverse events were recorded either as a volunteered statement by the patient at each visit, and if not volunteered then by specific inquiry by the attending physician. The nature of any adverse effects, including severity, time of appearance, duration and whether they could be attributed to the treatment, was recorded in case record forms. The event was carefully monitored to determine whether it corrected itself or if palliative medication was required.

Follow-up. Patients were asked to visit for follow-up examination every 2 weeks, at which time new scores were recorded by the investigators in a similar manner to the initial examination. At the same time, an inquiry was made with respect to the occurrence of adverse effects, which were noted accordingly. In the case of dropouts, attempts were made to determine the reason for dropping out of the study through telephone calls, letters or personal visits.

End point. The end point was reached when the patient completed 12 weeks of treatment or when disabling adverse effects required discontinuance of treatment, or the patient dropped out of the study at his/her own will.

Data analysis. An analysis of three major symptoms, including sneezing, rhinorrhea and nasal congestion, was conducted with respect to percentage improvement. Similarly, a comparison was made with regard to improvement of each patient based on the sum of all scores or total nasal symptom score (TNSS)(22) at weeks 0, 6 and 12. For example, if a patient's score was 60 for sneezing, 70 for rhinorrhea and 80 for nasal congestion, which reduced to was 30, 20 and 40, respectively, at week 6, then a TNSS of 210 was considered to be reduced to 90. Percentage improvement was calculated on the basis of reduction with respect to the initial scores using the following formula:

$$\% \text{ improvement} = \frac{(\text{initial score} - \text{final score})}{\text{initial score}} \times 100$$

$$\text{In the above example: } \frac{210 - 90}{210} \times 100 = 57.14\%$$

Sufficient clinical improvement was considered only if there was a reduction of more than 40% (23) in TNSS. The mean scores of individual symptoms and TNSS at weeks 0, 6 and 12 were also compared. The values obtained from the objective parameters were analyzed in a similar manner and were compared at weeks 0, 6 and 12.

Statistical analysis. To compare the means of two independent groups, the independent sample

(Student's) t-test was employed. When variances of the groups under comparison were heterogeneous, the t-test was used with modified degrees of freedom (df). Comparison of homogeneity of variances was determined using the two-tailed F_{max} test. To compare the means of dependent variables whose values were obtained from the same individual at different time points, repeated measures multivariate analysis of variance (RMANOVA) was used, with Pillai's trace as the statistical criterion. For the data analysis of STT, Wilcoxon's matched-pairs signed-ranks test was applied (24).

Results

Of the 374 patients, in the open-label group, 352 patients successfully completed the 12-week trial period, while 22 patients dropped out. In the double-blind, placebo-controlled study, 151 of the 171 patients completed the trial, out of which 79 received the test supplement, 72 received the placebo and 20 patients dropped out. Analysis of the data was conducted only on patients who completed the 12-week trial period.

Effect on individual symptoms

Open label. The assessment was performed on the three major symptoms (sneezing, rhinorrhea and nasal congestion) and TNSS (Table V). The calculation was done by aggregating scores of all the patients for each of the symptoms at weeks 0, 6 and 12. The mean score of each symptom was thus calculated. In comparison with the average scores at the point of entry (week 0), statistically significant decreases in sneezing scores at weeks 6 and 12 of treatment were observed in the treatment group [$F(2,350) = 1089.18$, $p < 0.001$]. Rhinorrhea and nasal congestion scores were significantly reduced at the end of weeks 6 and 12 of treatment relative to baseline, [$F(2,350) = 913.10$, $p < 0.001$] and [$F(2,350) = 478.37$, $p < 0.001$], respectively. Total nasal symptoms significantly decreased following treatment at the end of week 6 (59%) and week 12 (78%), indicating a significant improvement of TNSS [$F(2,350) = 1406.53$, $p < 0.001$].

Tables VIA and VIB show the percentage of patients reporting improvement in their symptoms at 6 and 12 weeks, respectively. At week 6, 71.06% of the patients reported an improvement of more than 40% in sneezing, 76.90% in rhinorrhea and 75.49% in nasal congestion. At week 12, 92.04%, 92.66% and 92.25% of the patients reported an improvement of more than 40% in sneezing, rhinorrhea and nasal congestion, respectively. At week 6, complete freedom from sneezing, rhinorrhea and nasal congestion was seen in 12.32%, 19.01% and 21.61% of the patients, respectively. At week 12, complete freedom from sneezing, rhinorrhea and nasal congestion was

Table V *Mean of the aggregate symptom scores of all patients*

Period	Sneezing (n = 351)[1]	Rhinorrhea (n = 342)[2]	Nasal congestion (n = 310)[3]	Total nasal mean ± SEM
Week 0	69.4 ± 1.13	67.64 ± 1.15	53.62 ± 1.51	190.67 ± 2.57
Week 6	30.11 ± 1.16*	26.19 ± 1.13*	21.11 ± 1.13*	77.41 ± 2.84**
Week 12	17.32 ± 0.95*	13.54 ± 0.90*	10.92 ± 0.87*	41.78 ± 2.30 **

Values represent mean ± SEM. *$p < 0.001$ *vs.* week 0 (baseline).**$p < 0.05$ *vs.* week 0 (baseline). [1]One patient did not have sneezing; [2]ten patients did not have rhinorrhea; [3]42 patients did not have nasal congestion.

Table VIA *Percentage relief of individual nasal symptoms in open-label study at 6 weeks*

	Sneezing		Rhinorrhea		Nasal congestion	
% Relief	*n* = 351	%	*n* = 342	%	*n* = 310	%
100	43	12.32	65	19.01	67	21.61
99-81	35	10.03	38	11.11	35	11.29
80-60	92	26.36	88	25.73	64	20.65
59 – 41	78	22.35	72	21.05	68	21.94
40-21	65	18.62	47	13.74	41	13.23
0-20	38	10.83	28	8.19	33	10.65
Worse	0	0.00	04	1.17	02	0.65

See Materials and methods section for details.

Table VIB *Percentage relief of individual nasal symptom in open-label study at 12 weeks*

	Sneezing		Rhinorrhea		Nasal congestion	
% Relief	*n* = 351	%	*n* = 342	%	*n* = 310	%
100	97	27.64	127	37.24	141	45.48
99-81	79	22.51	78	22.87	53	17.10
80-60	97	27.64	76	22.29	60	19.35
59-41	50	14.25	35	10.26	32	10.32
40-21	13	3.70	11	3.23	14	4.52
0-20	12	3.42	13	3.80	09	2.90
Worse	03	0.85	02	0.59	01	0.32

See "Materials and methods" section for details.

seen in 27.64%, 37.24% and 45.48% of the patients, respectively.

Overall improvement

Open label. Table VII shows the percentage of patients reporting improvement in their TNSS, as a percentage of improvement, at 6 and 12 weeks, respectively. At week 6, 78% of the patients reported an improvement of more than 40% in overall symptoms, while at week 12, 94% of the patients reported an improvement of more than 40% in overall symptoms, which were statistically significant from the baseline at both 6 and 12 weeks.

Randomized, double-blind, placebo-controlled studies. There were no significant differences in symptoms between the treatment and the placebo groups at baseline (week 0). At the end of weeks 6 and 12, patients in the treatment group showed a significant decrease in sneezing scores (68% and 74%), rhinorrhea (71% and 79%) and nasal congestion (66% and 73%) compared to placebo sneezing scores (51% and 55%), rhinorrhea (56% and 56%) and nasal congestion (56% and 58%). Although both treatment and placebo groups had significant decrease in sneezing and rhinorrhea symptoms, at the end of weeks 6 and 12 the treatment group had a significantly greater reduction in visual analogue scores than the placebo group (sneezing [$F(2,148) = 6.18$,

Table VII *Percentage of patients with an improvement in total nasal symptom scores (open label, n = 352)*

	Week 6		Week 12	
% Relief	*n*	%	*n*	%
100	18	5	64	18
99-81	75	21	129	37
80-60	86	24	103	29
59 - 41	94	28	34	10
40-21	51	14	13	4
0-20	26	7	8	2
Worse	2	1	1	0

See "Materials and methods" section for details.

$p < 0.003$], rhinorrhea [$F(2,148) = 8.26, p < 0.001$]) (Table VIII). Nasal congestion reduced significantly in both groups at weeks 6 and 12. At the end of 6 and 12 weeks, TNSS were significantly improved in the treatment group (68% and 75%) as compared to placebo (54% and 56%), and reduction of TNSS from the baseline to weeks 6 and 12 were significantly greater in the treatment group as compared to the placebo group [$F(2,148) = 6.57, p < 0.002$]) (Table VIII).

Overall improvement

Eighty-four percent of patients treated with PF and 75% of patients treated with placebo reported an overall improvement of more than 40% at the end of week 6, while 92% of patients taking PF reported an improvement of more than 40% at the end of 12 weeks *versus* 65% of those taking the placebo (Table IX).

Assessment of objective parameters for efficacy

Absolute eosinophil count. Compared with baseline count, absolute eosinophil count at week 12 was significantly reduced following treatment with PF (Table X).

Serum IgE. There was no consistent pattern of change in serum IgE levels as the values estimated at the end of 12 weeks of treatment showed both increases and decreases.

Mucociliary clearance time (MCT). The STT demonstrated the MCT in 30 patients before and after treatment in both the nostrils. In patients taking PF,

Table VIII *Mean of the aggregate symptom scores of all patients*

	Sneezing		Rhinorrhea		Nasal congestion		Total nasal score	
Period	Placebo	PF	Placebo	PF	Placebo (*n* = 72)	PF (*n* = 79)	Placebo	PF
Week 0	55.28 ± 2.55	58.29 ± 1.92	54.79 ± 2.31	58.42 ± 2.14	47.92 ± 2.51*	46.71 ± 2.68	157.99 ± 5.55	163.42 ± 5.02
Week 6	27.01 ± 2.52	18.48 ± 2.43[a]	24.37 ± 2.08	17.03 ± 2.36[b]	21.04 ± 2.05*	16.01 ± 2.3	72.43 ± 5.86	51.52 ± 6.30*
Week 12	25.07 ± 2.66	14.87 ± 2.50[a]	24.24 ± 2.51	12.47 ± 2.39[b]	20.35 ± 2.27	12.78 ± 2.22	69.65 ± 6.71	40.13 ± 6.35*

See "Materials and methods" section for details. Values represent mean ± SEM. PF = Polyherbal formulation; [a]$p < 0.003$ *vs.* placebo; *$p < 0.002$ *vs.* placebo: [b]$p < 0.001$ *vs.* placebo.

Table IX *Percentage of patients with overall improvement in total nasal symptom scores (double blind, n = 151)*

	Week 6				Week 12			
	Placebo		PF		Placebo		PF	
% Relief	*n*	%	*n*	%	*n*	%	*n*	%
100	6	8	14	18	9	13	24	30
99-81	6	8	20	25	10	14	25	33
80-60	22	31	23	30	18	24	12	15
59–41	20	28	9	11	10	14	11	14
40-21	8	11	7	9	10	14	1	1
0-20	10	14	5	6	15	21	5	6
Worse	0	0	1	1	0	0	1	1

See "Materials and methods" section for details.

mean values of 79.35 ± 17.0 at week 0 were reduced to 75.65 ± 30.5 at week 6 and were further reduced to 32.5 ± 6.7 at week 12 (Table XI). The MCT decreased in 21 patients, increased in seven patients and remained unchanged in two patients.

Peak expiratory flow rate (PEFR). PEFR values in 30 patients ranged from 250 to 700 l/min with an average value of 451 at week 0. At week 6 the average value increased to 491 and at week 12 the average was 486 (Table XI). There was a decrease in nine patients, an increase in 17 patients and no change in four patients.

Peak nasal flow rate (PNFR). Mean PNFR scores increased from 119.3 l/min at week 0 to 128.4 at week 6 and 156 at week 12 (Table XI). The values increased in 23 patients, deceased in five patients and remained unchanged in two patients.

Table X *Effect of treatment on absolute eosinophil count (open label, n = 174)*

Period	Absolute eosinophil count (cells/HPF)
Week 0	516 ± 33
Week 12	373 ± 24*

See "Materials and methods" section for details. Values represent mean ± SEM. *$p < 0.001$ *vs.* week 0 (baseline).

Skin prick test. Tests were conducted on 47 patients, who were found to be allergic with grades ranging from 1+ to 3+. Allergens included house dust, mites, Parthenium, cotton dust and Zeamays, *etc.* Twenty-three patients in the PF group had a positive skin prick test, whereas 24 patients, including one dropout, had a positive skin prick test in the placebo group.

Adverse events. All adverse events, whether volunteered or inquired about, were investigated in detail and are listed in Table XII. In all these instances the symptoms were mild, transient and spontaneously resolved.

Dropouts. Any patient who did not complete the full 12-week study period was considered a dropout. These patients were not included for analysis of improvement or of adverse effects. The total number of dropouts was 42, of which 35 were in the treatment group (*n* = 466) and seven were in the placebo group (*n* = 79). In the treatment group, six patients

Table XI *Effect of treatment on objective parameters*

Period	MCT (sec) (n = 30)	PEFR (l/min) (n = 30)	PNFR (l/min) (n = 30)
Week 0	79.35 ± 17.0	451 ± 18	119.3 ± 11.0
Week 6	75.65 ± 30.5	491 ± 31	128.4 ± 18.0
Week 12	32.5 ± 6.7*	487 ± 22	156.0 ± 13.0*

See "Materials and methods" section for details. Values represent mean ± SEM. *$p < 0.05$ compared with week 0 (baseline). MCT = mucociliary clearance time; PEFR = peak expiratory flow rate; PNFR = peak nasal flow rate.

Table XII *Incidence of adverse events*

Adverse events	PF (n = 431)		Placebo (n = 72)	
	n	%	n	%
I. Gastrointestinal:				
Gastritis	15	3.5	1	1.4
Epigastric discomfort	3	0.7	1	1.4
Vomiting	4	0.9	0	0
Burning in epigastric region	5	1.2	0	0
Dryness of mouth	2	0.5	0	0
Retrosternal discomfort	1	0.2	0	0
Constipation	2	0.5	0	0
Anorexia	1	0.2	0	0
Increased appetite	0	0.0	1	1.4
Abdominal pain/discomfort	1	0.2	0	0
Loose motions	4	0.9	0	0
Stomach discomfort	1	0.2	0	0
Total gastrointestinal	39	9.1	3	4.2
II. Other:				
Weakness/fatigue	2	0.5	3	4.2
Itching of the skin	2	0.5	1	1.4
Congestion of chest	2	0.5	0	0
Headache	2	0.5	0	0
Skin rashes	3	0.7	0	0
Increased frequency of urination	1	0.2	0	0
Burning micturition	1	0.2	0	0
Total other	13	3.0	4	5.6
Total adverse events	52	12.1	7	9.7

PF = polyherbal formulation.

dropped out due to no improvement, one dropped out due to abdominal discomfort, one due to skin rash, one due to increased symptoms and two,due to complete improvement. The remaining 24 dropped out for unknown reasons. In the placebo group, six dropped out due to no improvement and one due to unknown reasons (Table XIII).

Biochemical analyses. As a part of the safety evaluation, laboratory tests were performed to evaluate hematology and renal and liver function prior to and after treatment. Although minor differences were observed in some of the parameters, these were within the routine laboratory ranges.

Discussion

Allergic rhinitis is a widely prevalent global health problem related to environmental allergens such as house dust mites, mite feces or animal dander (1, 2). When a patient suffers from allergic rhinitis at a particular time of the year, it usually indicates seasonal allergy. When symptoms exist throughout the year it is called perennial. The clinical condition is considered chronic when repeated episodes of symptoms are experienced. The initial reaction takes place between an allergen and mast cells, and typical histopathology of nasal mucosa would draw profound submucosal edema with eosinophilic infiltration. In more severe symptoms, basophils release histamine and spasmogenic substances such as leukotrienes. Cytokines are humoral factors released by immunologically active cells. They are secreted when cells are activated by antigens and other cytokines such as interleukins (IL)-2, IL-4, IL-9 and IL-10, which act as promoters of IgE production. IL-4 activates monocytes/macrophages and IL-2, IL-3, IL-5 bring about eosinophilia (25).

Typically, patients resort to several therapeutic and nontherapeutic measures to alleviate their symptoms. Among available conventional therapies, antihistamines, mast cell stabilizers and corticosteroids are widely used. However, adverse effects, low tolerance, dependence and limitations in carrying out normal day-to-day activities tend to restrict their use (8). Natural products validated by scientific research and adequate safety studies have received special attention. However, the apparent disadvantage is that herbal remedies have a longer onset of action than conventional drugs.

Our earlier studies have demonstrated the broad spectrum safety of this novel PF in a number of toxicity studies including acute oral, subacute, subchronic, Ames' bacterial reverse mutation assay, reproductive and teratogenic toxicity studies (13). PF also exhibited potent efficacy in mast cell stabilization, lipoxygenase and hyaluronidase inhibition, as well as antihistaminic and antispasmodic activities (10). Recent studies also demonstrated novel antioxidant and anti-inflammatory potential of PF in a number of *in vitro* and *in vivo* models (11, 12).

A phase I clinical trial conducted in healthy huma volunteers (n = 20) established the tolerability of P up to a dose of 1,980 mg/day (14). A pilot efficac study (n = 17) and questionnaire-based study (n = 3(revealed the efficacy of PF in allergic rhinitis patient at a dose of 1,320 mg/day for 6-12 weeks. In add

Table XIII *Dropouts*

	Polyherbal formulation		Placebo	
Reason	*n*	%	*n*	%
Unknown reasons	24	5.15	1	1.26
100% improvement	2	0.43	0	0
No improvement	6	1.29	6	7.59
Increase in symptoms	1	0.21	0	0
Abdominal discomfort	1	0.21	0	0
Skin rashes	1	0.21	0	0
Total dropouts	35	7.50	7	8.85

tion, no significant adverse effects were observed. A phase II double-blind, placebo-controlled study ($n = 48$) was then conducted, which established the efficacy of PF in allergic rhinitis patients at a dose of 1,320 mg/day for 12 weeks. Importantly, PF produced significant improvement in quality of life parameters compared with placebo. PF was also well tolerated in this study (14). At this stage it was desirable to carry out clinical assessment in a larger number of subjects under open and blind conditions.

The diagnosis was established based on the presence of at least two out of three significant nasal symptoms (sneezing, rhinorrhea and/or nasal congestion) (23) for a minimum period of one year and that at the time of presentation the symptoms had been present for at least 15 days. Most of these patients were undergoing treatment with antihistamines and/or intranasal corticosteroids.

In the open trial, a significant improvement in sneezing, rhinorrhea and nasal congestion of more than 40% was observed (23) at the end of 6 weeks indicated by reduction in total nasal symptoms. At the end of 12 weeks, a further significant improvement in all symptoms was observed.

In the double-blind, placebo-controlled trials, patients treated with PF showed a significant improvement in these individual symptoms at 6 weeks, which further improved at 12 weeks. In the placebo group, improvements of 40% or greater were observed at the end of 6 weeks, but no further improvement was seen thereafter. At 6 and 12 weeks, overall total nasal scores were reduced in 83.5% and 91.1% of the patients taking PF compared with 75% and 65.2% of the patients taking a placebo. The remarkable reduction in symptoms in the early part of treatment, with PF and the placebo group, could have been due to better clinical care. However, the fact that the improvement continued to occur consistently beyond 6 weeks in patients taking PF indicates a pharmacodynamic effect.

IgE is identified as a reaginic antibody produced in response to antigens. Asthma, allergic rhinitis and atopic dermatitis are accompanied by elevated levels of IgE (1). The concept that allergen-specific IgE initiates acute allergic airway symptoms and promotes an ongoing allergic response has stimulated the development of therapeutic agents to block the interaction of IgE with its high affinity receptor (4). Preclinical data on animals demonstrated that PF reduced the synthesis of reaginic antibodies in response to horse serum (26). Hence, it was considered worthwhile to investigate the immunological mechanisms of PF. Further studies utilizing specific IgE estimations could be more conclusive than the present study. However, after treatment the absolute eosinophil count was significantly reduced in the PF-treated group.

STT objectively measures the interaction between ciliary movement and removal of mucus secretion from the nose (5). Altered ciliary clearance time has been identified using STT in asthma (27), chronic bronchitis (28), bronchiectasis (29) and sinusitis (30). An association of increased STT in allergic rhinitis and its improvement with mometasone nasal spray has been established (5). This parameter appeared useful as an objective parameter for assessing clinical improvement; hence, it was considered worthwhile to investigate the effects of PF on STT. PF improved the MCT from an average of 79.4 seconds to 75.7 and 32.5 seconds at weeks 6 and 12, respectively. Thus, PF seems to have a significant effect on mucociliary clearance, resulting in increased ciliary movement and faster removal of mucus secretion.

It has been observed that the patients with allergic rhinitis have an altered PEFR and bronchial hyperresponsiveness, even in the absence of symptoms of bronchial asthma. PEFR was measured at week 0 and at every follow-up visit using a Wright's peak flow meter. The average PEFR ranged from 450 to 700 l/min with a mean value of 451 l/min, except in one patient whose PEFR was 250 l/min. In patients taking PF, PEFR was improved to 491 l/min at week 6 and 486 l/min at week 12; however, neither of these observations was statistically significant. Since the majority of patients included in the trial had a normal PEFR, PF probably had not caused further enhancement of this function.

Nasal obstruction is one of the important features of perennial rhinitis, which is rarely relieved by conventional therapy. Various methods have been used to measure nasal congestion, including anterior rhinomanometry and electronic peak flow meter (31, 32). We attempted a simpler method using a Wright's flow meter with modification, which showed that PF improved PNFR from 119.3 to 156 l/min, which was statistically significant ($p < 0.05$). This supports the observations of symptomatic improvement in nasal congestion.

Skin prick test provides a useful tool for determining allergy history and provides a visual demonstration for the patient; it was used for the selection of 55 patients in one center to observe the therapeutic response in the positive or negative groups. The test was not repeated at 12 weeks and, as the time for reversal may be at least 6-12 months in some cases, no reversal takes place for years.

Conventional antihistamines are typically accompanied by adverse effects, mainly sedation. During 12 weeks of treatment, not a single patient on PF showed any such features. Although there were some minor variations from their baseline values, all biochemical and hematological parameters of PF-treated patients were within normal limits.

Management with pharmacological agents acting as specific end-organ antagonists for allergic rhinitis include antihistamines, mast cell stabilizers or anti-inflammatory agents, which are very effective for some patients but which may lead to the adverse effects of sedation or gastrointestinal distress. Local or oral α-adrenergic decongestants lead to rebound vasodilatation and chronic rhinitis with prolonged use. Local or topical steroids lead to adrenal suppression and such agents can be used only for a short period for temporary amelioration of symptoms. An herbal agent, butterbur *(Petasites hybridus)*, a plant native to Europe, Northern Africa and South Western Asia, has traditionally been used in bronchial asthma due to its inhibition of biosynthesis of leukotrienes (33). Butterbur was tested against cetirizine to assess its tolerability and efficacy in seasonal allergic rhinitis (34). Both the agents were found to be similar in efficacy and tolerance. However, the cetirizine group had sedative effects in 8 out of 12 patients, although the drug is considered as a nonsedating antihistamine (34).

Taking all aspects of the multicenter clinical trial, this novel PF (Aller-7/NR-A2) effectively demonstrates its clinical usefulness in ameliorating symptoms of allergic rhinitis and in providing complete relief to a section of patients. It appears to be well tolerated by most patients. Only a minority experienced some adverse effects, which were trivial and self-limiting.

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